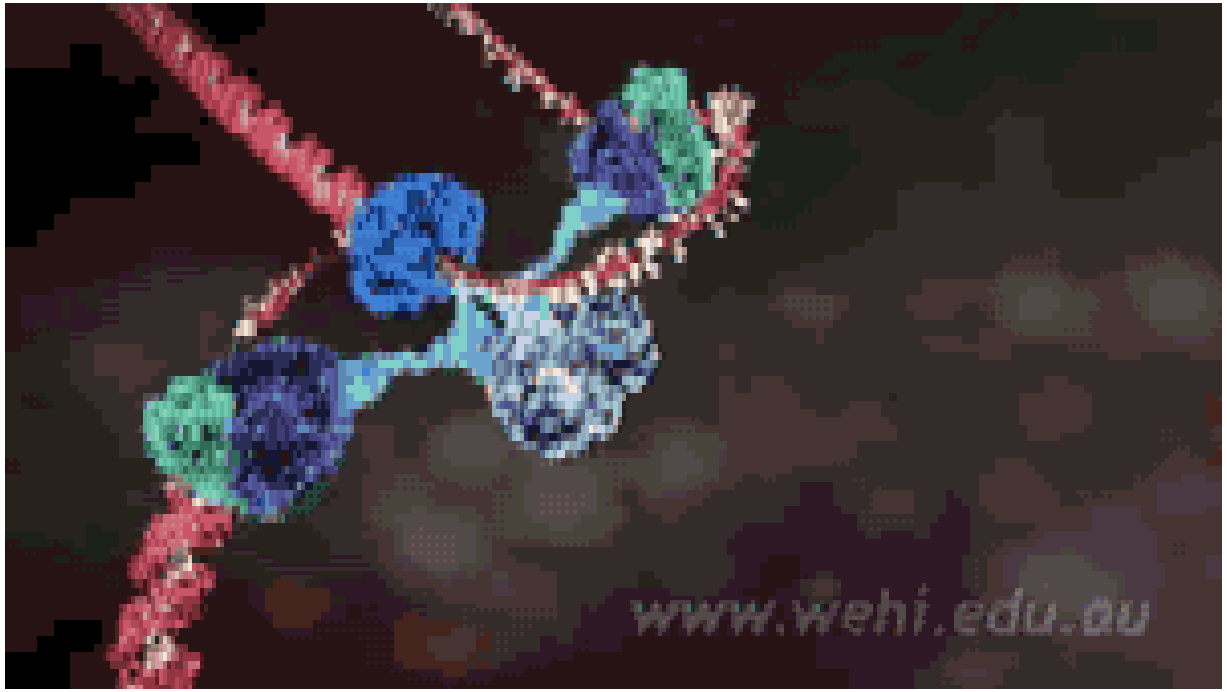
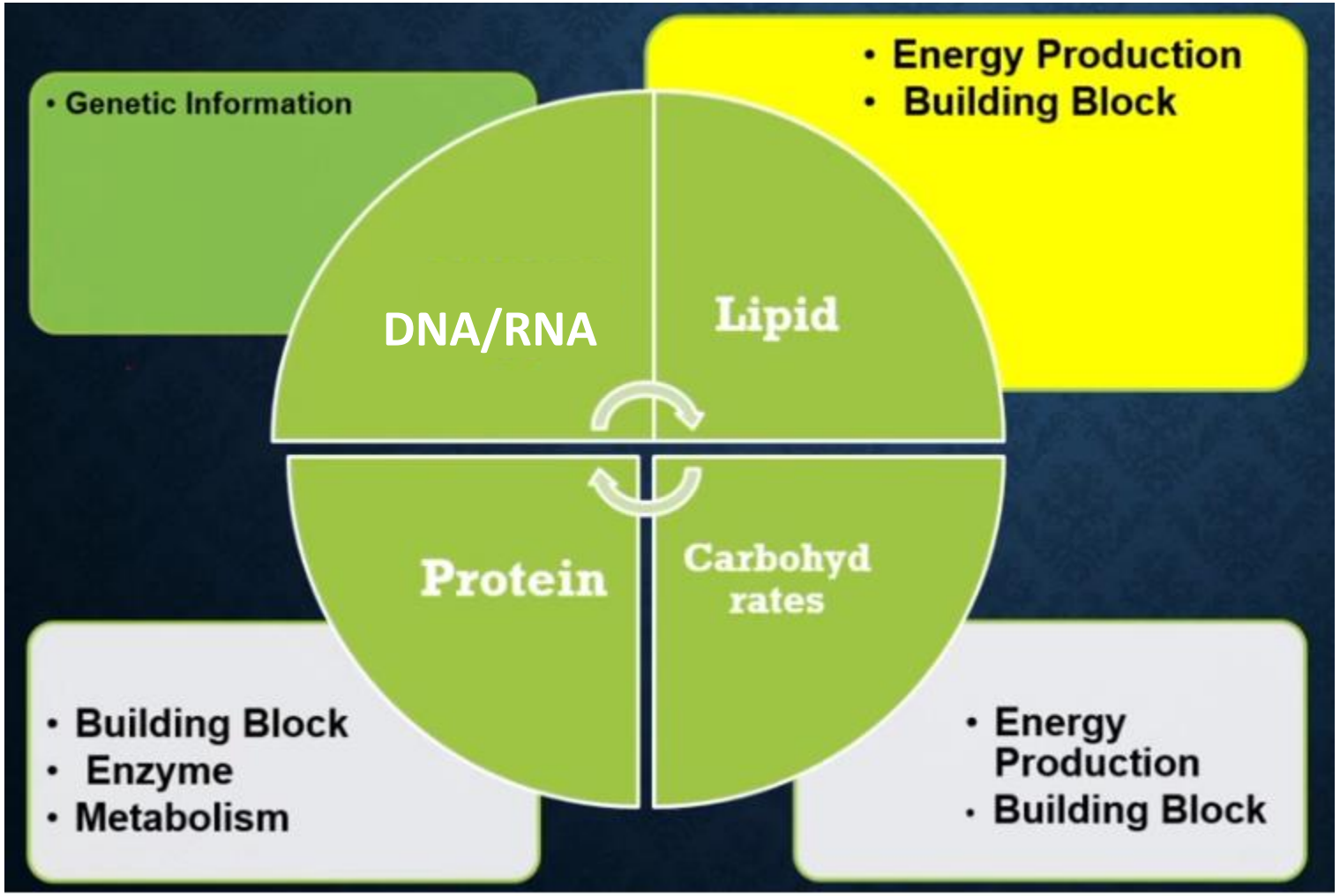
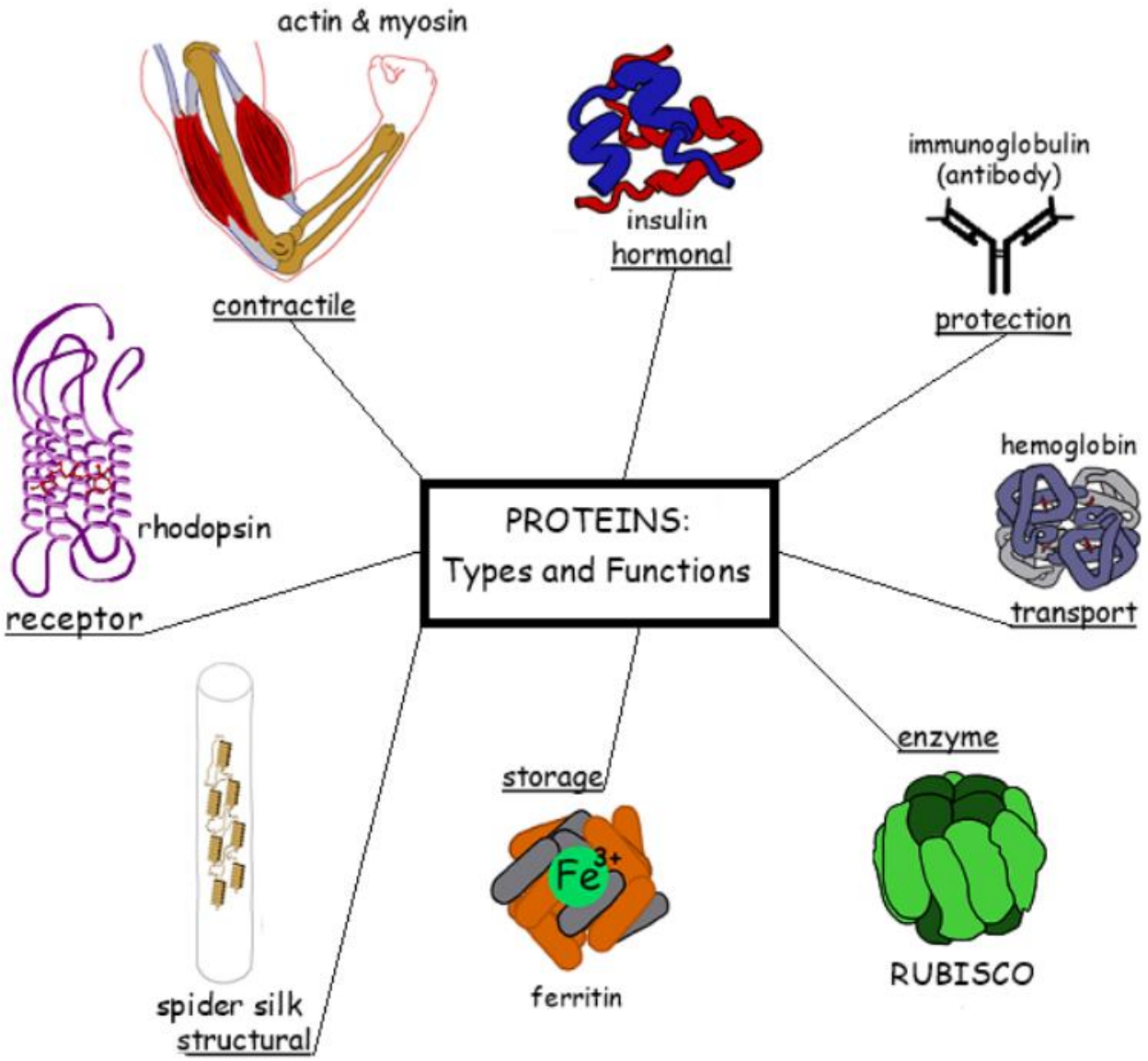
Biomolecules - Proteins

Lecture 10




• Genetic Information
• Energy Production
• Building Block
DNA/RNA
Lipid
Protein
Carbohyd rates
• Building Block
• Enzyme
• Metabolism
• Energy Production
• Building Block

The composition of biomolecules

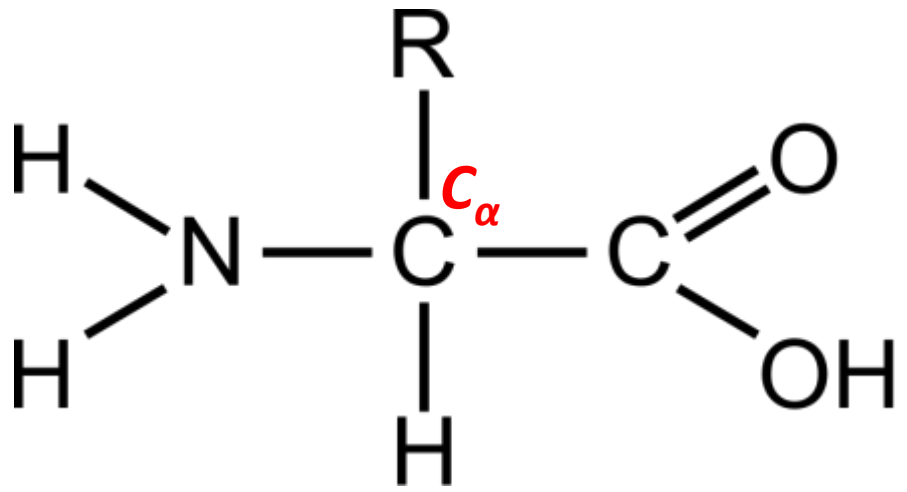


- Proteins account for more than 50% of the dry mass of most cells.
- Protein functions include structural support, storage, transport, cellular communications, movement, and defense against foreign substances.



Structure of Amino acids

Each amino acid has 4 different groups attatched to α-carbon (which is C atom next to COOH). These 4 groups are : amino group, COOH group, Hydrogen atom and side chain(R).



Amino acids differ in their properties due to differing side chains, called R groups

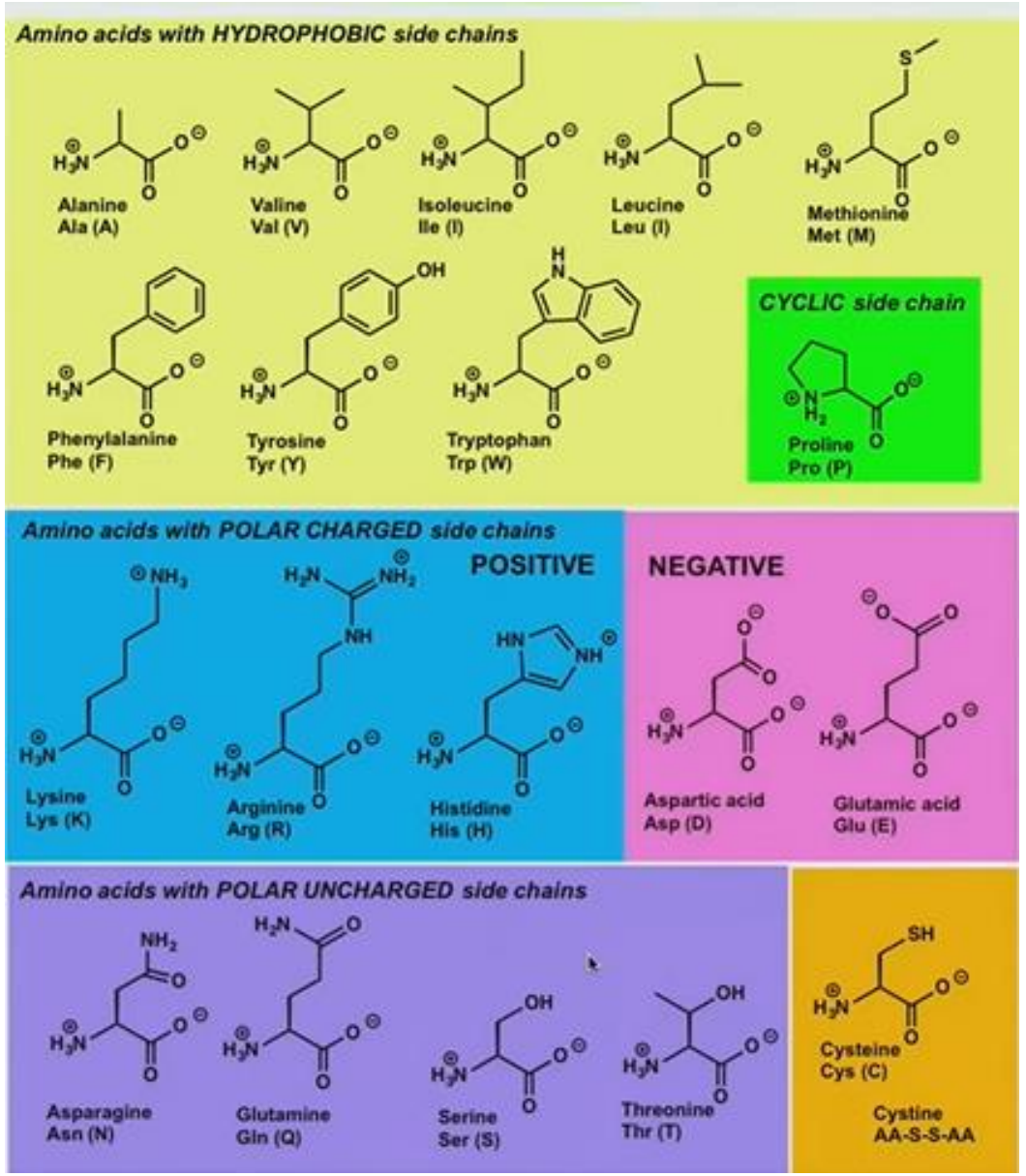
Amino acids, peptides and proteins

- The physical and chemical properties of the side chain (R group) determine the unique characteristics of a particular amino acid
- Amino acids assemble into unique sequence of linear polymer to form Primary structure.

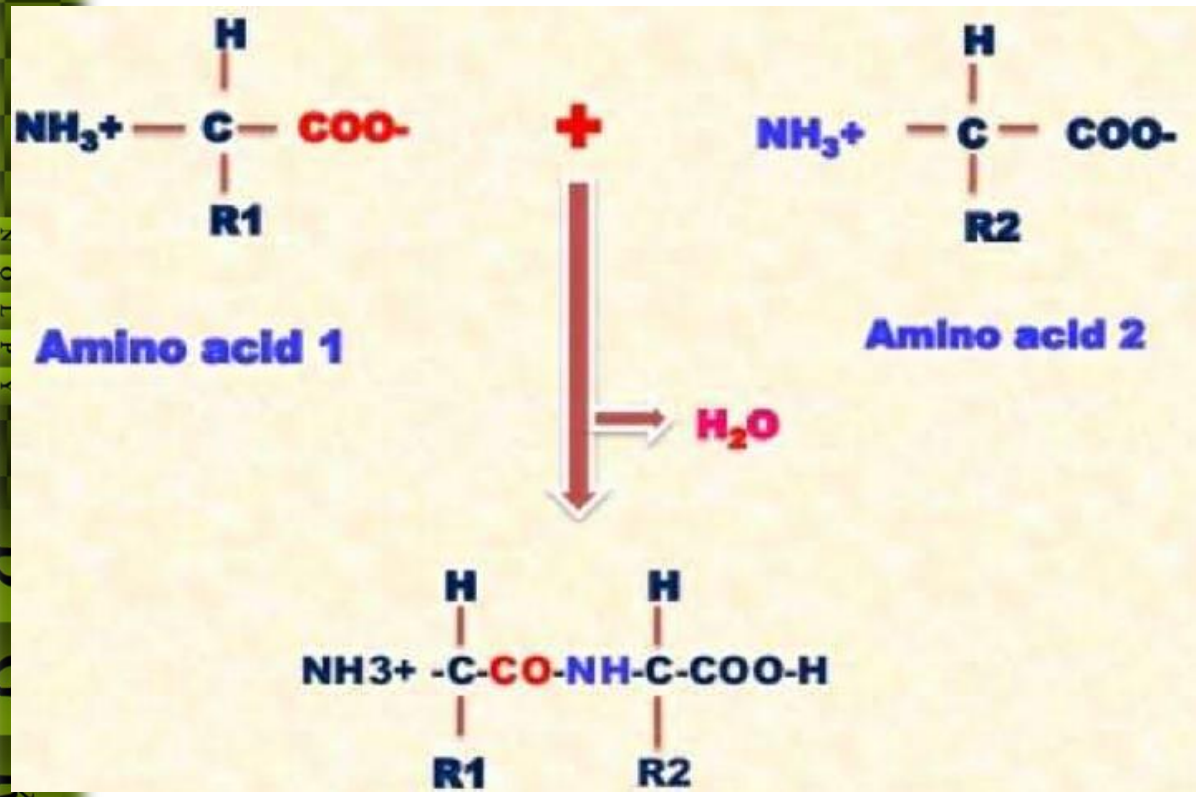
- The primary sequence determines the fold of the protein in 3-dimension.

- Thermodynamically most stable structure exists.

- The fold determines the function of proteins.



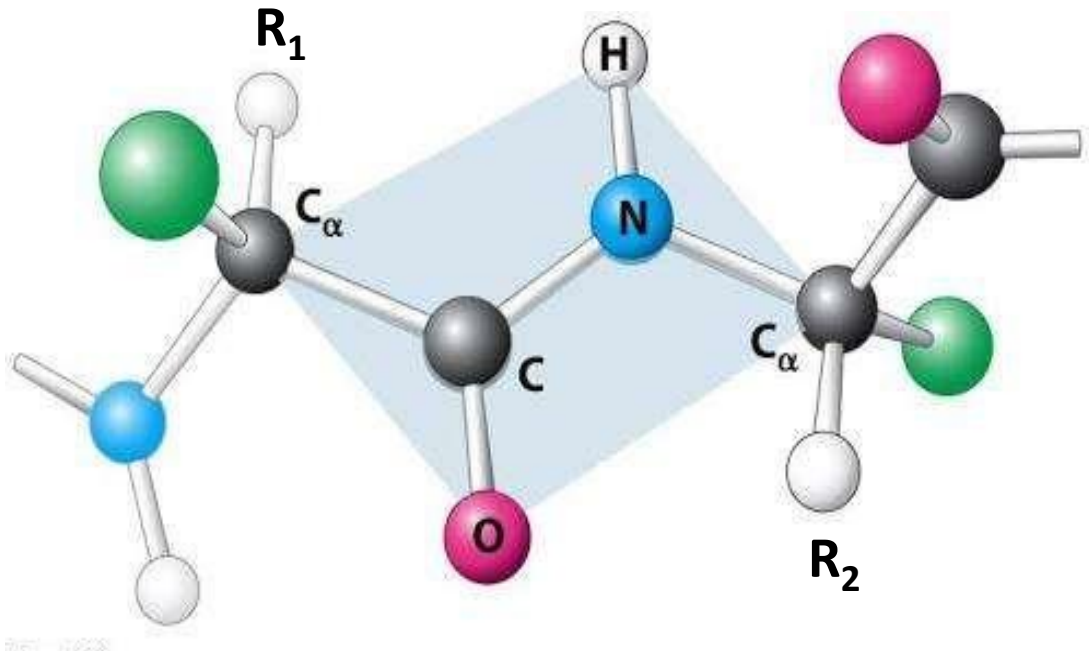
Peptide bond formation

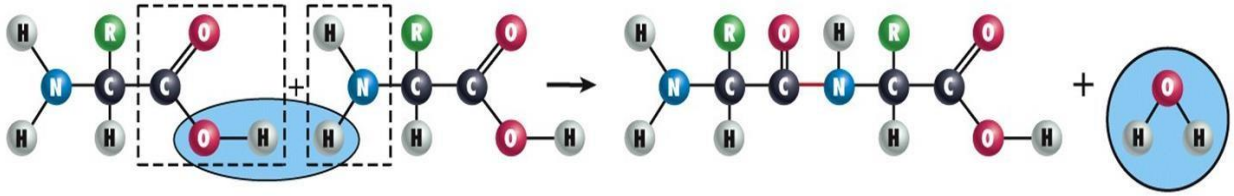


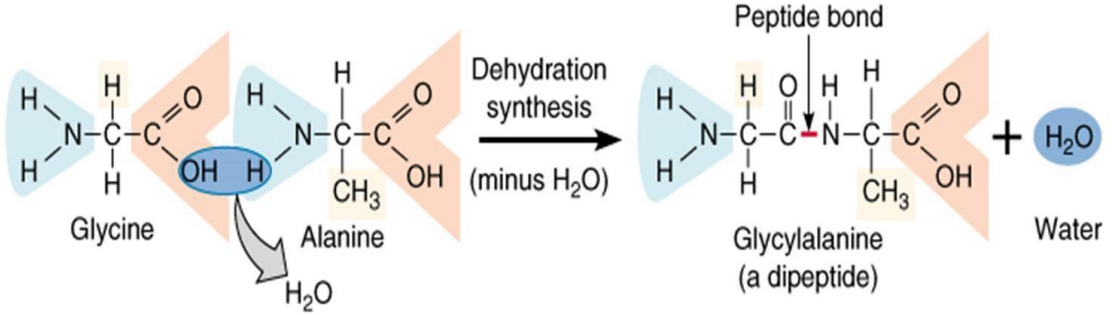
- Amino acids are linked by peptide bonds.
- A polypeptide is a polymer of amino acids.
- Polypeptides range in length from a few to more than a thousand monomers.
- Each polypeptide has a unique linear sequence of amino acids, with a carboxyl end (C-terminus) and an amino end (N-terminus).

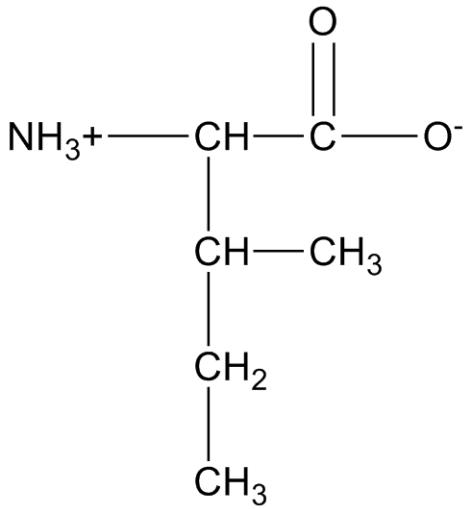
Peptide bond formation

- α- carboxyl group of one amino acid (with side chain R1) forms a covalent peptide bond with α- amino group of another amino acid (with side chain R2)by removal of a molecule of water.
- Formation of dipeptide results in release of water molecule due to condensation.
- Repetition of the process leads to the formation of polypeptide chain.
- AA chains less than 40 do not form a well-defined structure and are usually called peptides

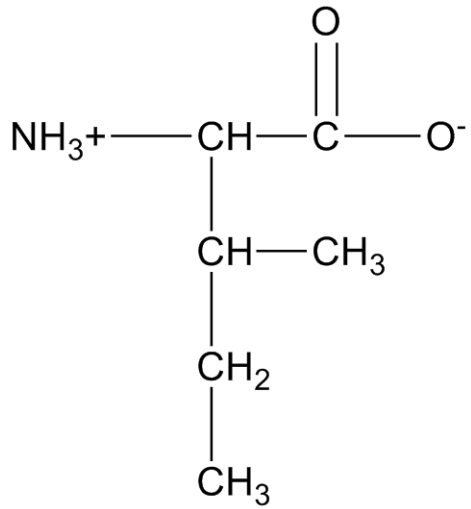




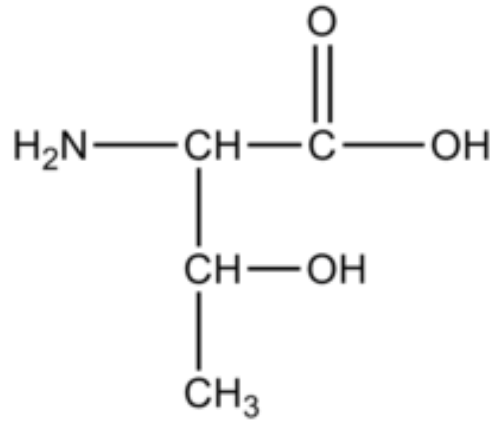




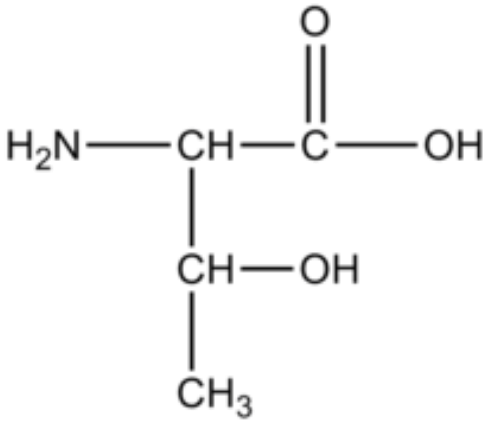
Isoleucine (I)



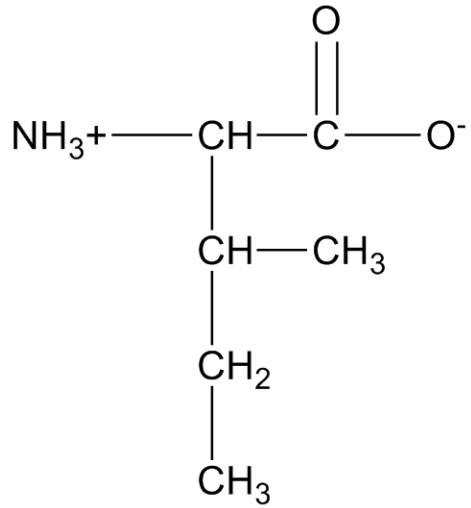
Isoleucine (I)



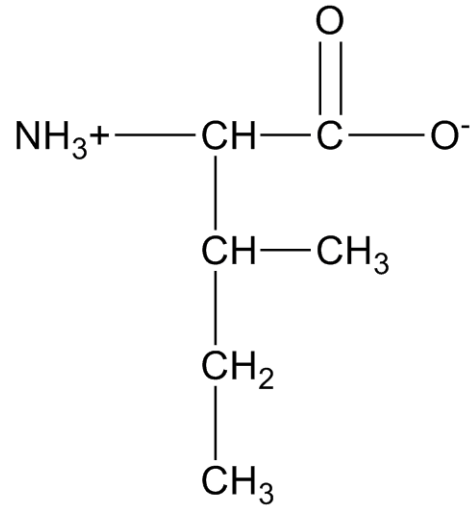
Threonine (T)



Threonine (T)

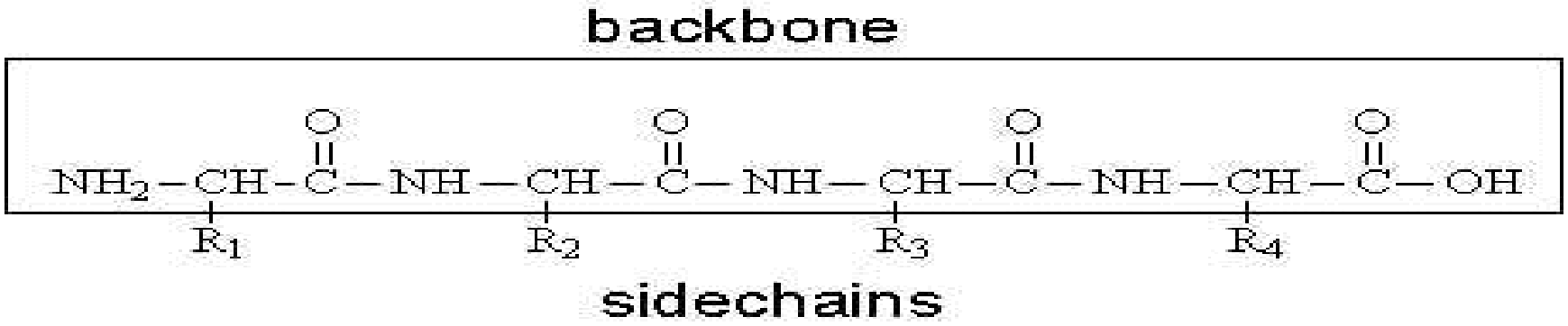


Isoleucine (I)



Isoleucine (I)

Peptide bond formation



- Polypeptide backbone is the repeating sequence of the N-C-C-N-C-C… in the peptide bond.
- The side chain or R group is not part of the backbone or the peptide bond.

Protein structure

Protein structure or fold is described as:

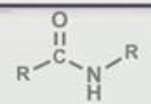
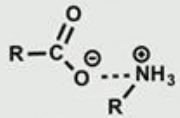
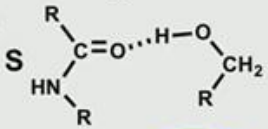
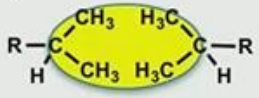
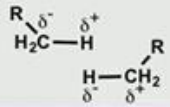
Primary sequence Based on covalent bonds

Secondary structure

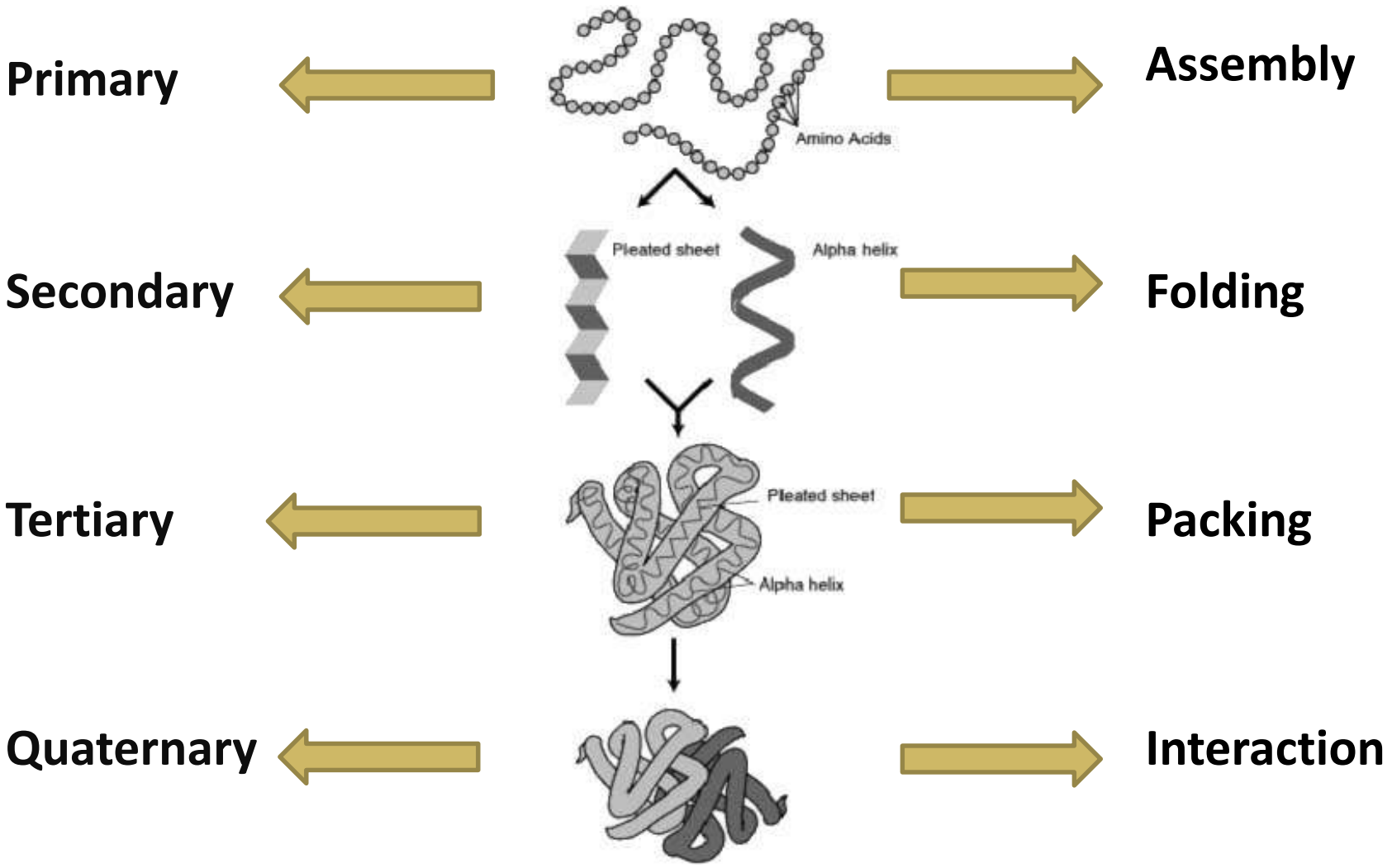
Tertiary structure

Quaternary structure

} Based on non-covalent interactions

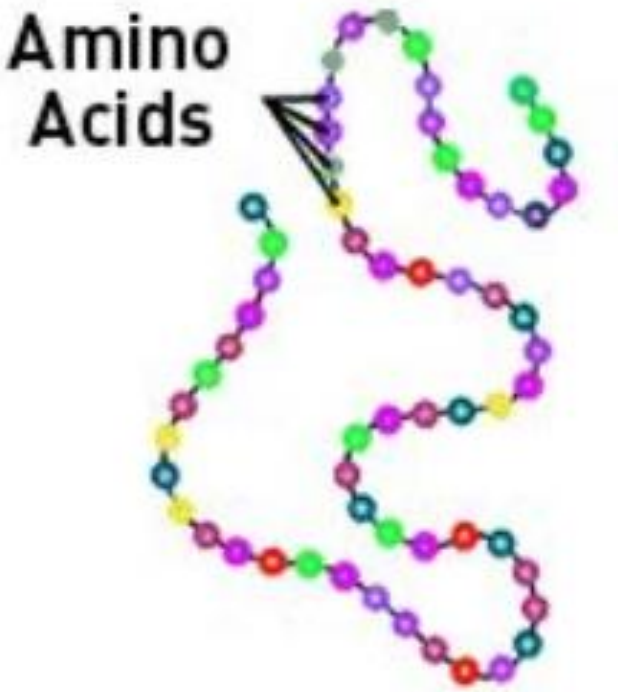
Name	Interaction	Structure	Bond Energy (kcal/mol)
Covalent	Shared electrons		50-100
Ionic (salt bridge, electrostatic)	Attraction of opposite charges		very variable
Hydrogen bond	Interaction of H on N, O, S with electron pair on O, N, S		3-7
Hydrophobic	Interaction of non-polar groups Exclusion of water		1-2
van der Waals	Interactions between weakly polarized bonds		1

Protein structure



Primary structure

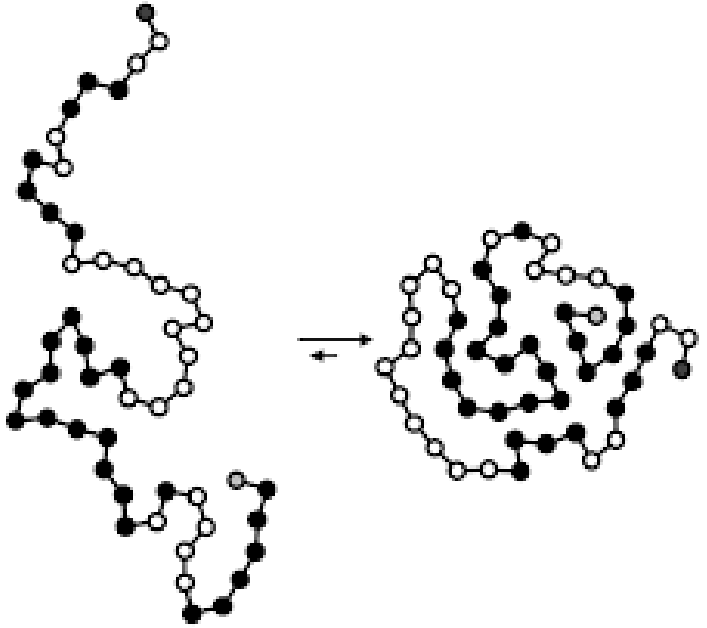
- The primary structure of protein refers to the sequence of amino acids present in the polypeptide chain.
- Amino acids are covalently linked by peptide bonds or covalent bonds.
- By convention the primary structure of protein starts from the amino terminal (N) end and ends in the carbonyl terminal (C) end.



Protein folding

- The peptide bond allows for rotation around it and therefore the protein can fold and orient the R groups in favorable positions.

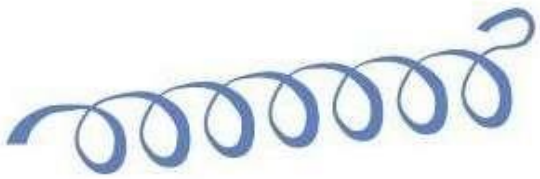
- Weak non-covalent interactions will hold the protein in its functional shape – these are weak and will take many to hold the shape.



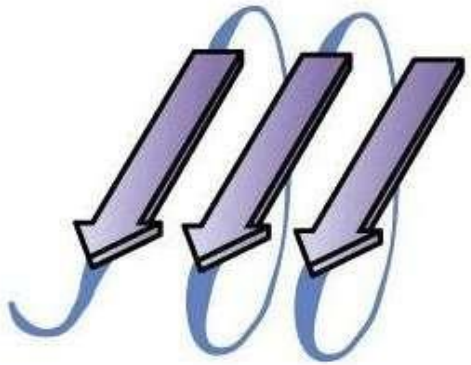
Secondary structure

- It is a local, regularly occurring structure in proteins and is mainly formed through hydrogen bonds between backbone atoms.

- Pauling & Corey studied the secondary structures and proposed 2 conformations
  - α helix
  - β sheets.



α helix



β sheet

Alpha Helix

- Right handed spiral structure.
- Side chain extend outwards.
- Stabilized by H bonding
- Amino acids per turn – 3.6
- Alpha helical segments, are found in many globular proteins like myoglobin,troponin C.
- Length ~12 residues and ~3 helical turns.

Ala, Leu, Met, Phe, Glu, Gln, His, Lys, Arg

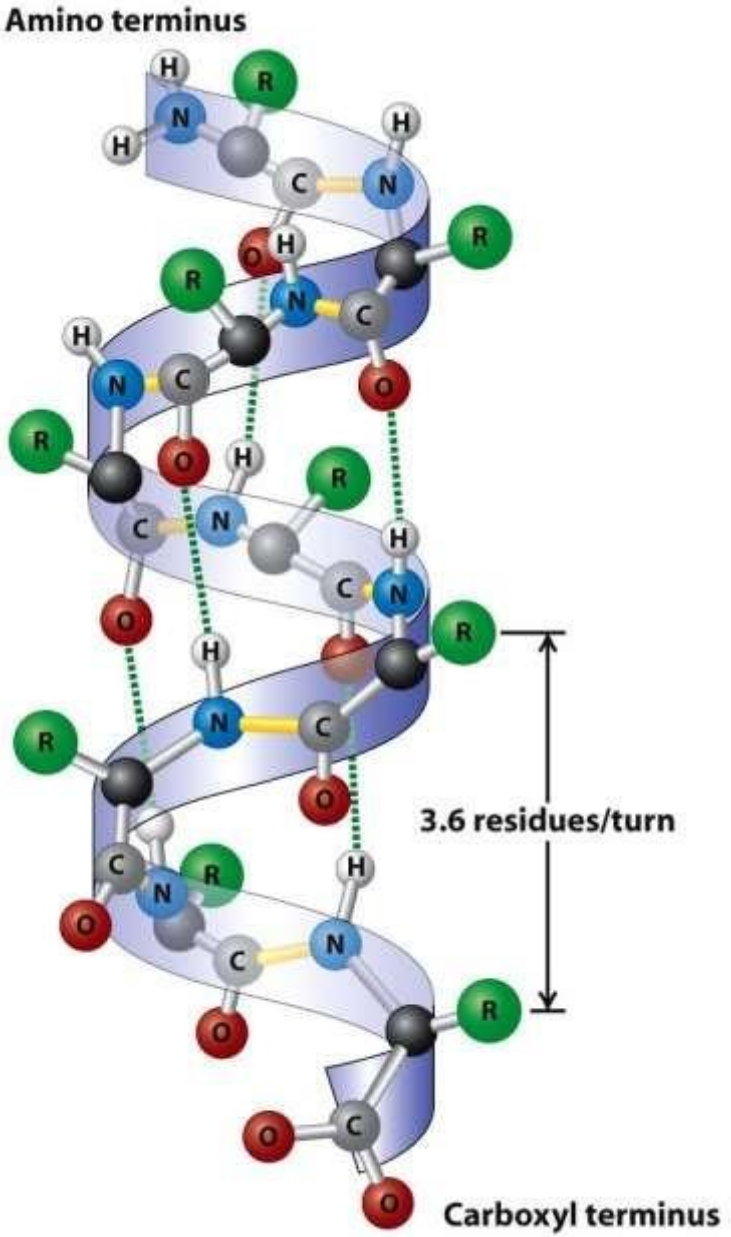
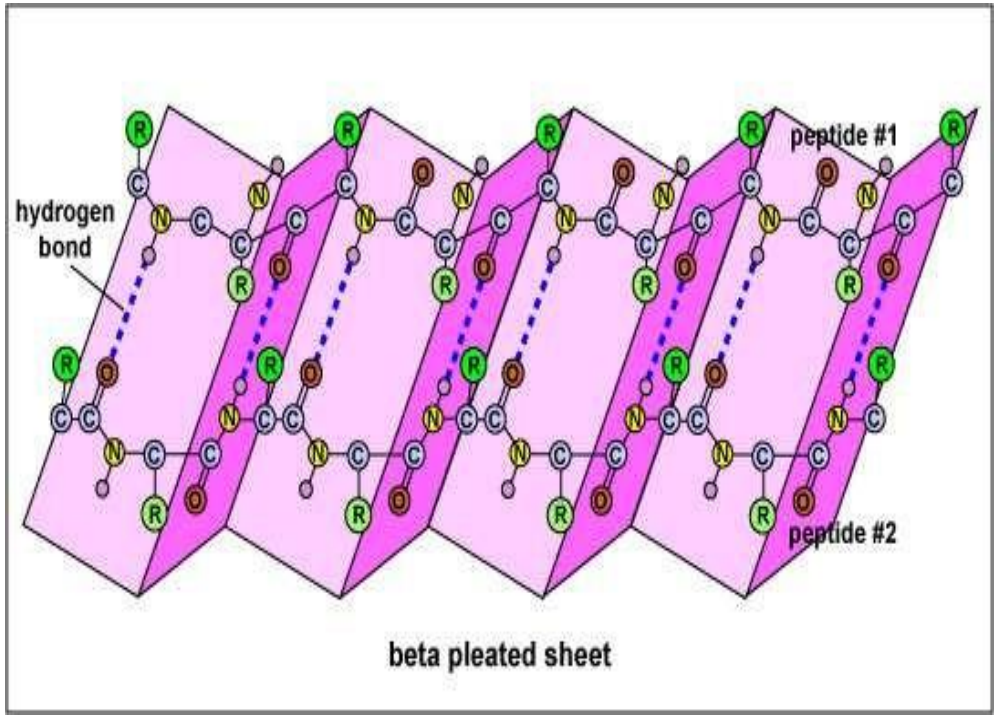
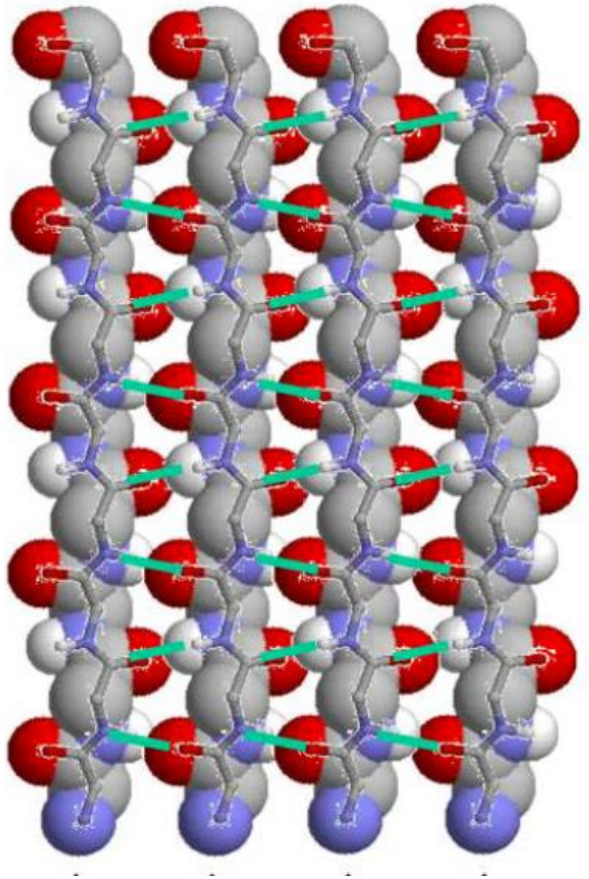


Figure 3-4
Molecular Cell Biology, Sixth Edition
© 2008 W. H. Freeman and Company

Beta sheet

- Formed when 2 or more polypeptides line up side by side.
- They are stabilized by hydrogen bond between N- H and carbonyl groups of adjacent chains.



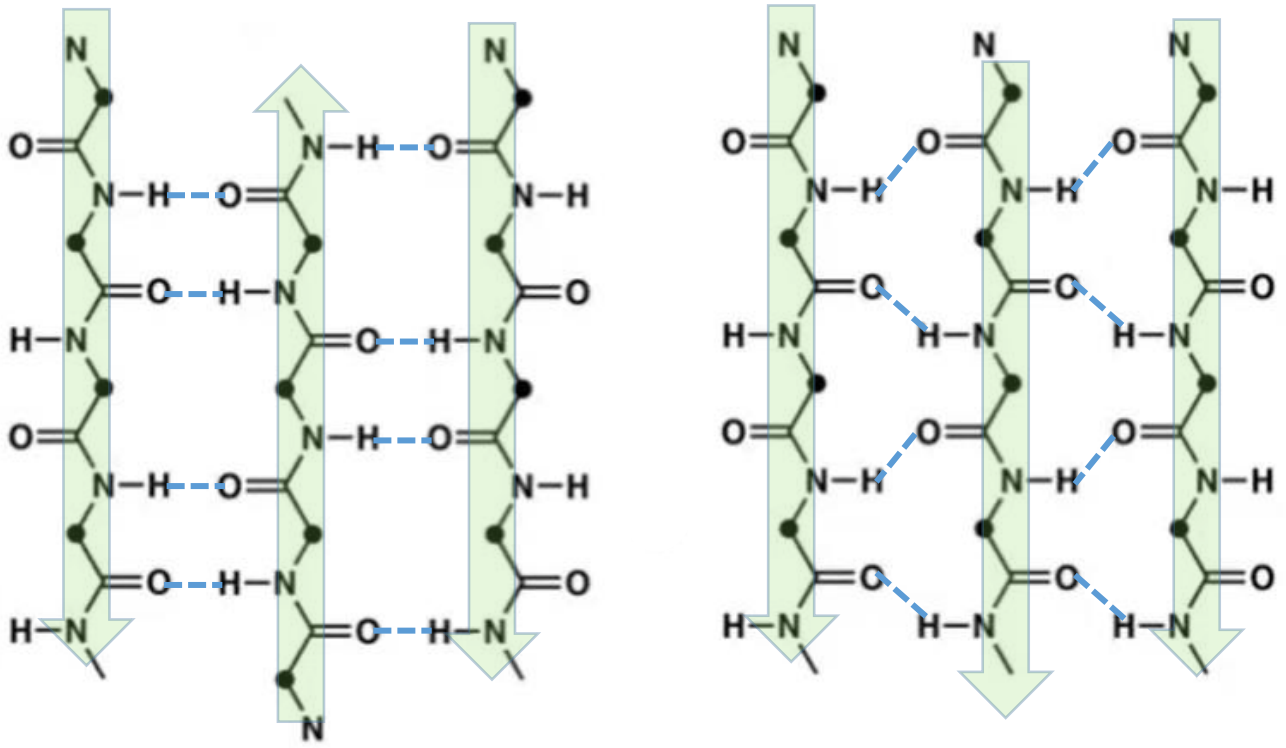


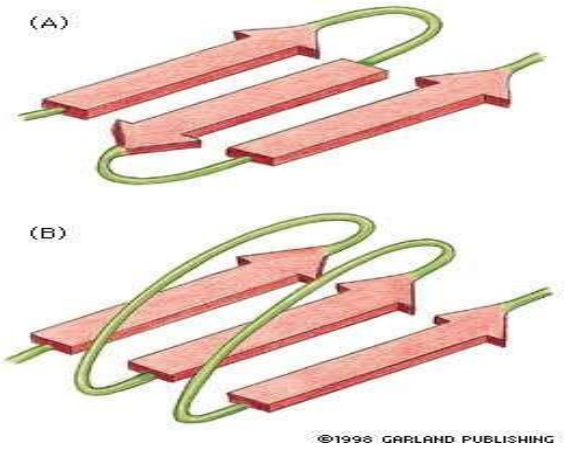
Side chains lie above and below the sheet.

Image: Michael Levit

Beta sheet

- Formed when 2 or more polypeptides line up side by side.
- They are stabilized by hydrogen bond between N- H and carbonyl groups of adjacent chains.

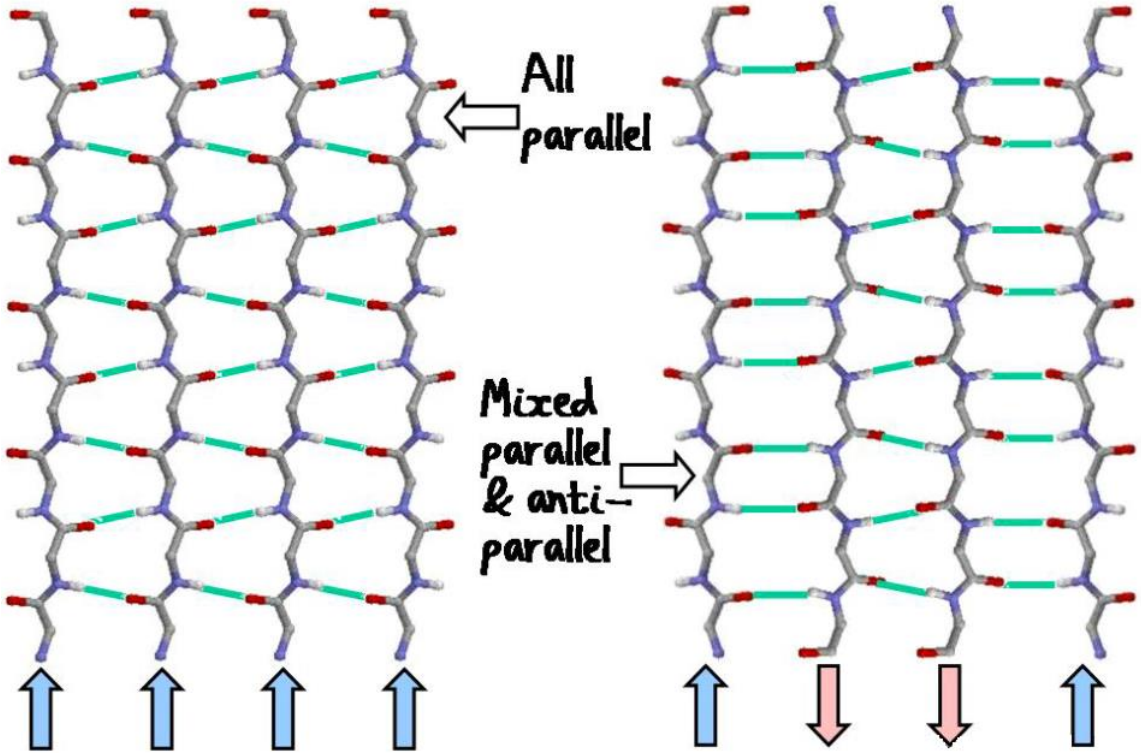


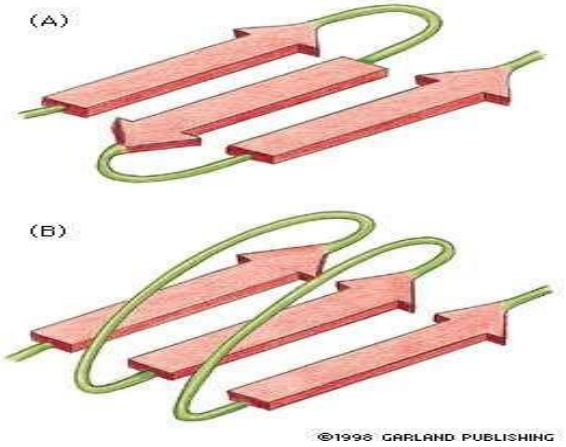


Tyr, Trp, Phe, Met, Ile, Val, Thr, Cys

Beta sheet

- Formed when 2 or more polypeptides line up side by side.
- They are stabilized by hydrogen bond between N- H and carbonyl groups of adjacent chains.





Tyr, Trp, Phe, Met, Ile, Val, Thr, Cys

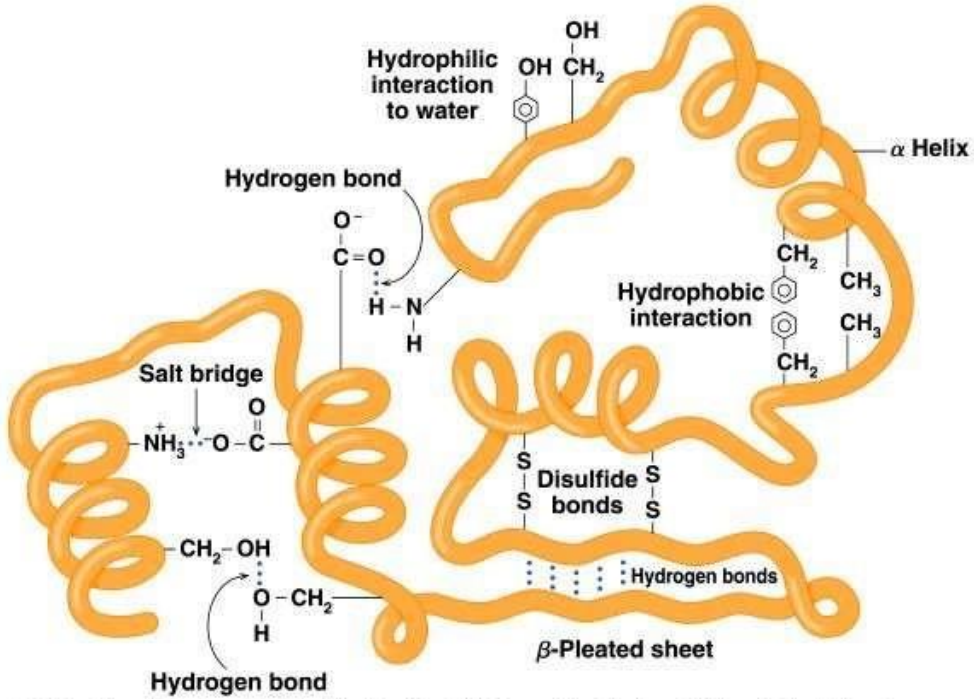
Tertiary structure of Proteins

The **tertiary structure** defines the specific overall **3-D shape** of the protein.

Tertiary structure is based on various types of interactions between the side-chains of the peptide chain

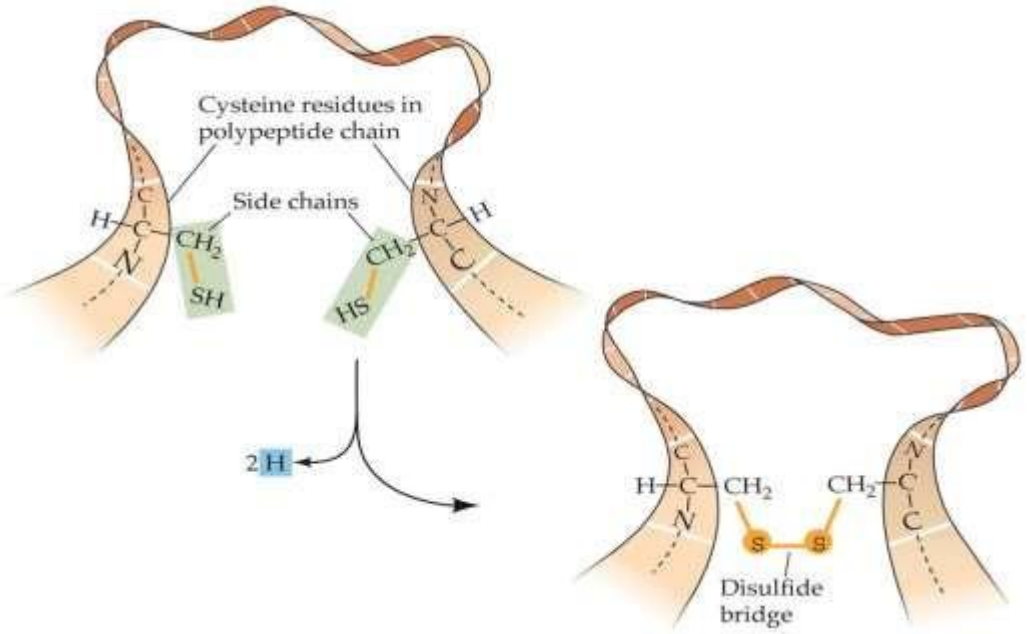
Interactions stabilizing tertiary structure :

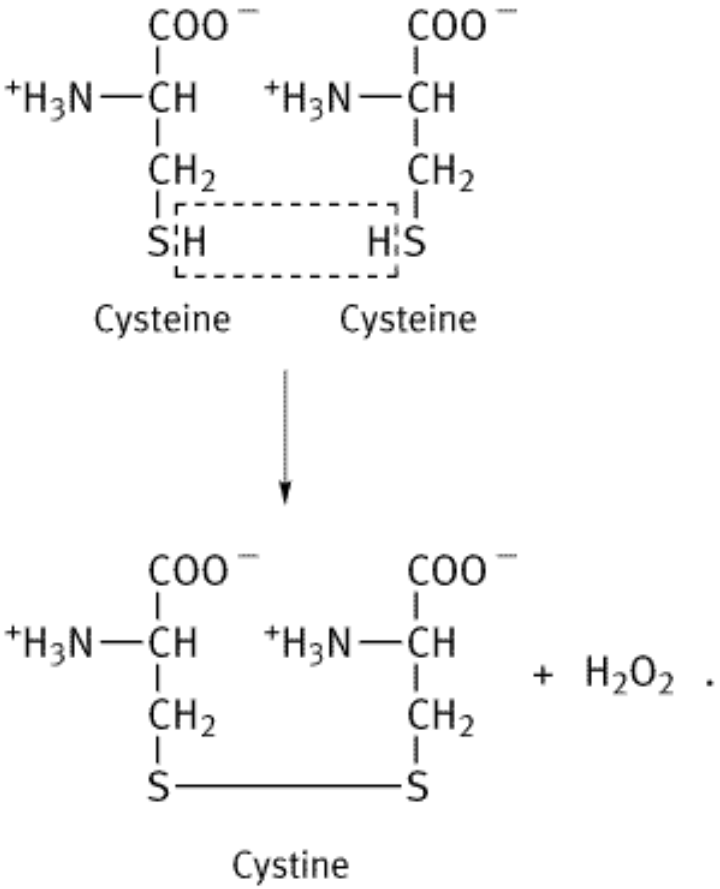
1. Disulfide bonds
2. Hydrophobic interactions
3. Hydrogen bonds
4. Ionic interactions
5. Vander Waals force



Timberlake, *General, Organic, and Biological Chemistry*. Copyright © Pearson Education Inc., publishing as Benjamin Cummings

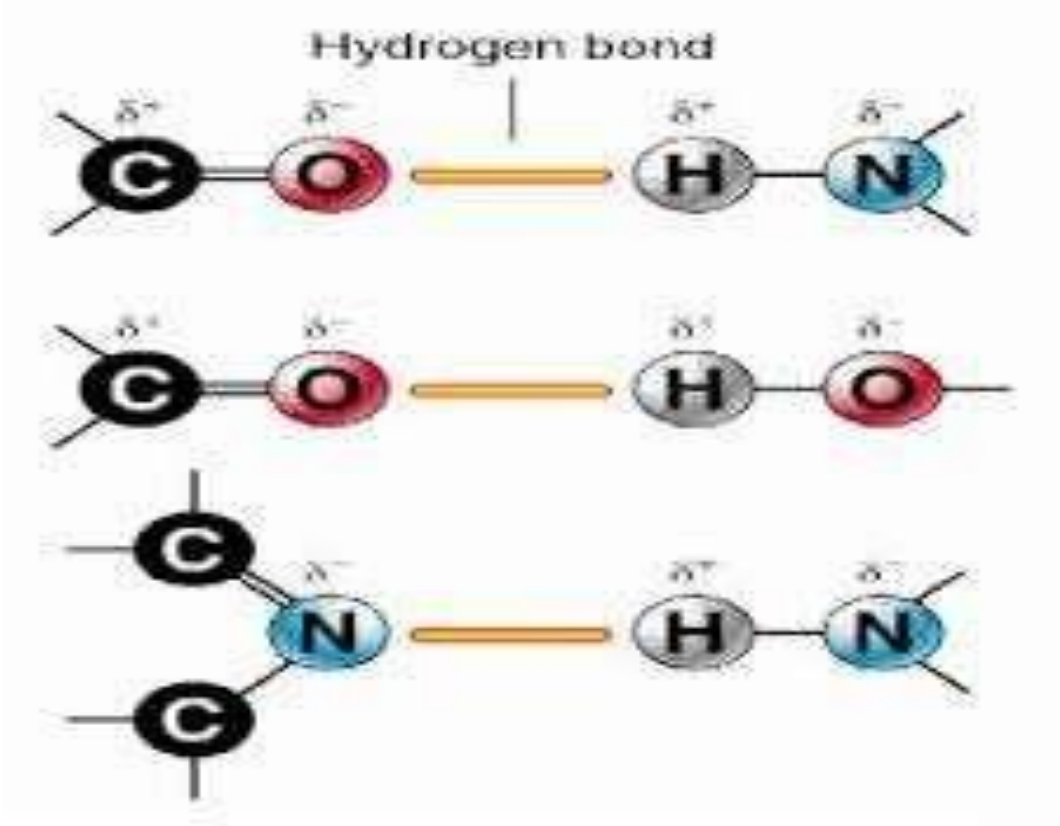
Tertiary structure – disulphide bonds





Covalent bond between sulfur atoms on two cysteine amino acids.

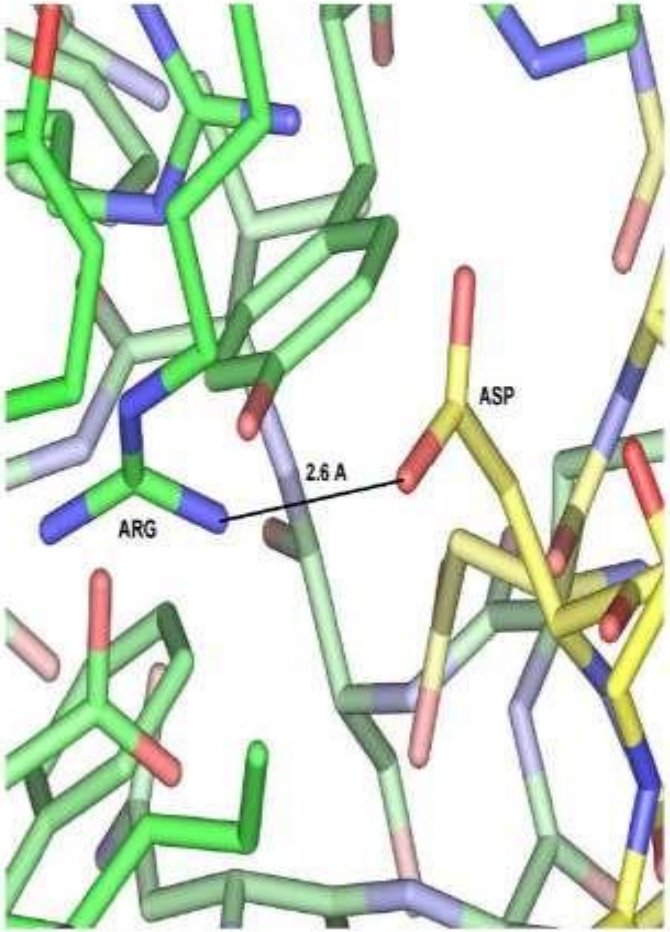
Tertiary structure – hydrogen bonds

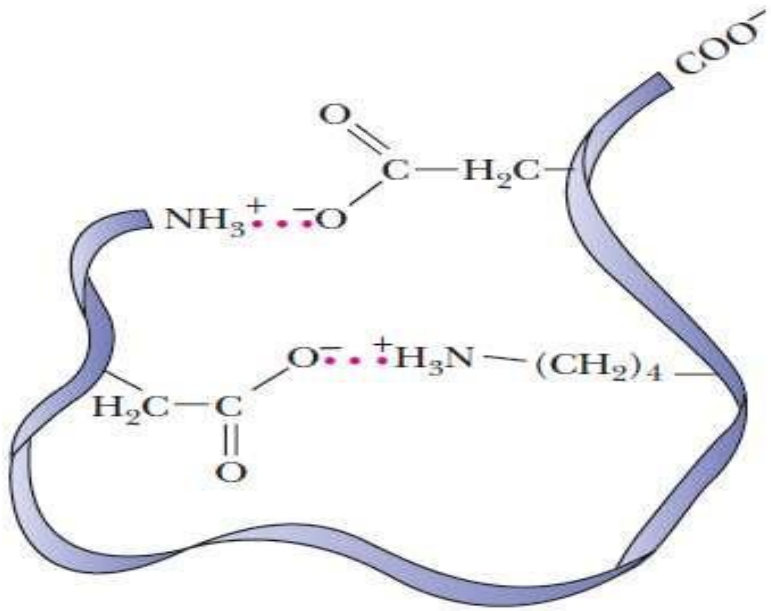


- H bonds are weak which allows to be broken and reformed easily.
- Allows structural change and produces ‘functional’molecules

Tertiary structure – salt bridge

- Salt bridges are electrostatic bonds between oppositely charged groups
- The strength is usually 4-7 kcal/mol

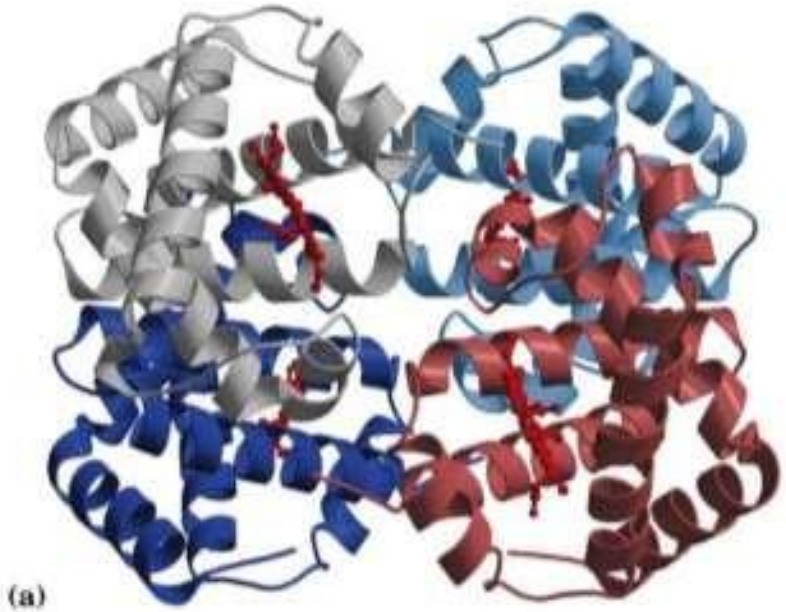




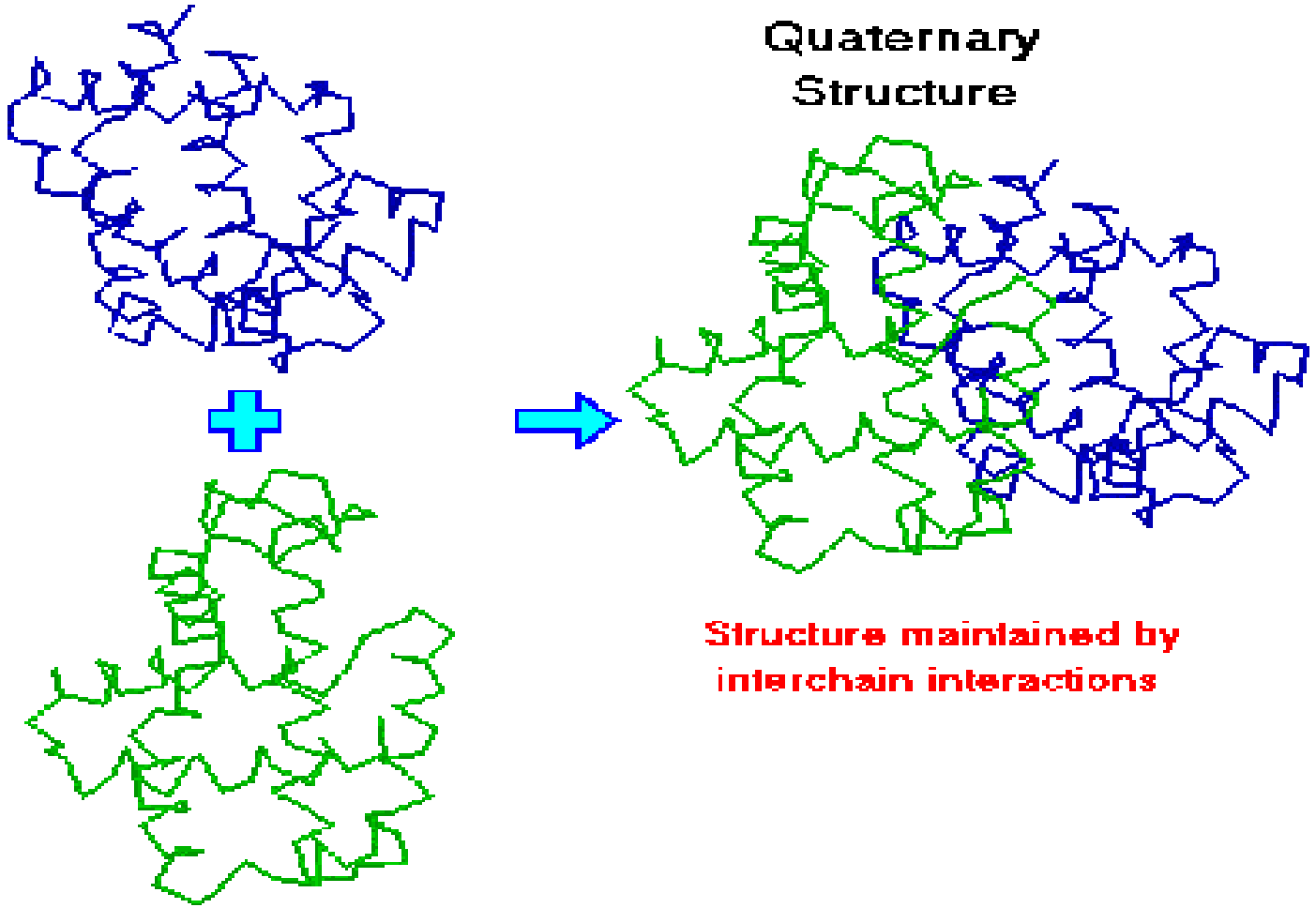
- Ions on R groups form salt bridges through ionic bonds.
- NH_3^+ and COO^- areas of the protein attract and form ionic bonds.

Quaternary structure of proteins

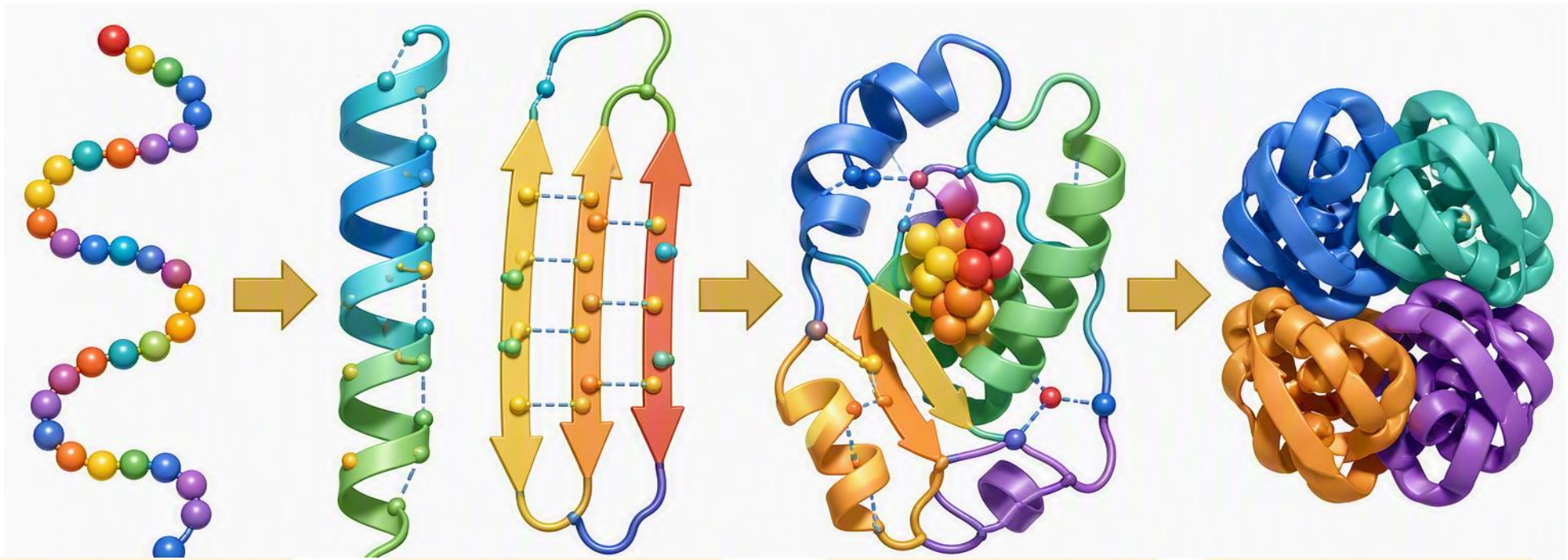
- The quaternary protein structure involves the clustering of several individual peptide or protein chains into a final specific shape.
- A variety of bonding interactions including hydrogen bonding, salt bridges, and disulfide bonds hold the various chains into a particular geometry.
- Two kinds of quaternary structures: both are multi-subunit proteins.
  - **Homodimer**: association between identical polypeptide chains.
  - **Heterodimer**: interactions between subunits of very different structures.
- The interactions within multi subunits are the same as that found in tertiary and secondary structures



Quaternary structure of proteins



Structural configuration of proteins



Primary Protein Structure
Sequence of a chain of amino acids

Covalent bonding

Secondary Protein Structure
Local folding of the polypeptide chain into helices or sheets

Non-covalent bonding in the backbone

Tertiary Protein Structure
three-dimensional folding pattern of a protein due to side chain interactions

Non-covalent bonding in the backbone and surrounding materials

Quaternary Protein Structure
protein consisting of more than one amino acid chain

Non-covalent bonding in the backbone and surrounding materials

The protein databank (PDB)

- http://www.rcsb.org/pdb/home/home.do
- A collection of (almost) all published experimental structures of biomacromolecules (e.g., proteins)
- Each identified by 4-character code (e.g., 1rcx)
- Currently ~120,000 structures. 90% of those are determined by x-ray crystallography.
- Browse it and look at some structures.

